Seeing Through Solid Rock: Finding a Hidden Chamber with Cosmic-Ray Muons

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Abstract

Muography images the interior of a dense object by counting cosmic-ray muons that survive passage through it. We built a reproducible Python simulation of a 230 m square-based limestone pyramid containing a 10 m radius spherical chamber, observed by a 0.25 m^2 detector with a $\pm 40^\circ$ field of view and 2° angular pixels. The simulation combines a measured cosmic-ray muon spectrum, exact range-energy energy loss, ray-marched opacity, expected counts, Poisson noise, and two distinct significance definitions. For the baseline model the expected signal in a fixed direction reaches three standard deviations (3σ) after 4.8 days and five standard deviations (5σ) after 13.4 days, thus the 30-day expected significance is 7.50σ, representing a deterministic benchmark for the stated model. Using the exact pixel solid angle lengthens the 5σ exposure to 15.4 days, a whole-image significance requires about 60 days, and a 4% error in the assumed rock density can fake a 5σ signal at 180 days. All four named parameter studies were completed, which include chamber radius, chamber depth, detector placement, and angular binning. A hypothetical Malaysian limestone hill with an 8 m cave would require about 9 days. The work shows how geometry, counting statistics, and systematics control the detectability of a subsurface void, and provides a reproducible template for teaching muon radiography.

Keywords: cosmic-ray muons, muography, Monte Carlo simulation, Poisson statistics, exposure optimisation.

1 Introduction

Every minute, roughly 10^4 cosmic-ray muons cross every square metre of ground at sea level. They are produced when high-energy cosmic rays strike nuclei in the upper atmosphere, creating pions that decay into muons. A muon is a heavy cousin of the electron which has the same charge, about 207 times the mass, and a rest lifetime of 2.2 μs. A naive calculation says that a muon travelling at nearly the speed of light should decay after about 660 m, so almost none should reach the ground

from the upper atmosphere. They arrive anyway because their lifetime is dilated by the relativistic factor γ, thus making cosmic-ray muons as one of the most direct everyday demonstrations of special relativity.

The same properties make muons a penetrating probe. In the minimum-ionising regime a charged particle loses about 2 MeV per gram per square centimetre, so a muon with tens of GeV can cross tens of metres of rock. A muon is heavy enough that the radiative loss which stops an electron is suppressed by a factor of order 207^2, and it does not feel the strong force that absorbs protons and alpha particles. The result is a natural X-ray machine the size of the sky, which means that a detector beside a mountain counts muons from each direction, and a line of sight with less material transmits more muons than the surrounding rock [1, 2].

Muography is an old idea. Alvarez et al. searched for hidden chambers in the Pyramid of Khafre in 1970 [3]. It became prominent in 2017 when the ScanPyramids collaboration reported a large void above the Grand Gallery of Khufu's pyramid, found independently with nuclear emulsion, scintillator and gas detectors [4]. Modern reviews describe applications from volcanoes to blast furnaces and nuclear reactors [1]. What is often missing from introductory treatments is a complete, reproducible end-to-end calculation that states explicitly how the exposure time is defined, how expected and observed significances differ, and how large the leading systematics are. That gap is the motivation for this work.

This paper answers a deliberately narrow question:

> *Under a simplified, fully specified simulation model, how many days of exposure does a 0.25 m^2 detector need before the expected chamber signal reaches 5σ?*

The objective is not to predict the performance of a real detector, nor to reproduce the ScanPyramids analysis. It is to build a transparent model in which every assumption can be inspected, to validate the geometry and statistics against hand calculations, and to quantify how the answer changes when one parameter at a time is varied. The result is a benchmark and a teaching tool, and its limitations are stated explicitly in Section 6.

2 Methodology

2.1 Approach and tools

The study is a computational experiment. A three-dimensional density model is ray-traced to obtain the opacity in 1681 viewing directions, the opacity is converted to expected muon counts, and the counts are sampled with a Poisson distribution. The approach is appropriate because the physical quantity of interest, which is the exposure at which a small density contrast becomes statistically significant, is not available analytically for a finite angular grid and a realistic target shape. A simulation also allows the same geometry to be run with and without the chamber, which is the reference needed to divide out the pyramid's shape.

The implementation uses Python 3.11.17, NumPy 2.4.6 [5], Matplotlib 3.11.2 [6], and Jupyter 1.1.1 [7]. The supplied physics is in `code/muography.py`, the analysis is in `code/student_analysis.py`, and all baseline values are in `code/project_config.py`. The reference random seed is 0 and the notebooks with a summary JSON file are listed in Appendix D.

2.2 Target geometry and detector

The target is a square-based pyramid with base 230 m, height 139 m, and density 2.5 $\mathrm{g\,cm^{-3}}$. Its half-width at height z is

$$w(z) = 115\left(1 - \frac{z}{139}\right)\ \mathrm{m}. \tag{1}$$

Table 1. Baseline simulation parameters, stored in `code/project_config.py`.

Parameter	Baseline	Units
Detector area	0.25	m^2
Detector position	$(0, -40, 3)$	m
Pyramid base / height	230 / 139	m
Limestone density	2.5	$\mathrm{g\,cm}^{-3}$
Chamber radius	10	m
Chamber centre	$(0, 0, 70)$	m
Angular field	± 40	degrees
Angular bin size	2	degrees
Ray step	0.25	m
Reference random seed	0	—

The chamber is a sphere of radius 10 m centred at $(0, 0, 70)$ m, filled with air at 1.2×10^{-3} $\mathrm{g\,cm}^{-3}$. The detector is a 0.25 m^2 square at $(0, -40, 3)$ m, with a $\pm 40^\circ$ field of view and $2^\circ \times 2^\circ$ pixels, giving $41 \times 41 = 1681$ directions. Table 1 lists the baseline parameters. The detector lies inside the 230 m footprint, and in the model this is interpreted as a detector in an accessible niche or passage at the base, and the rock between that niche and the outside is not modelled.

2.3 Ray tracing and expected counts

For every direction the opacity is

$$\varrho = \int \rho(\mathbf{r})\, d\ell, \tag{2}$$

evaluated by marching along the ray in 0.25 m steps and summing $\rho\,\Delta\ell$, with $\Delta\ell$ converted from metres to centimetres. The camera is a tangent plane, as a direction is specified by two pointing angles,

$$\mathbf{d} = \frac{(\tan a_x, \tan a_y, 1)}{\sqrt{1 + \tan^2 a_x + \tan^2 a_y}}, \tag{3}$$

so that $\cos\theta = 1/\sqrt{1 + \tan^2 a_x + \tan^2 a_y}$. This parameterisation keeps the angular grid rectangular and makes the zenith angle a direct function of the two angles.

The energy-loss rate is modelled by

$$\left\langle -\frac{dE}{dX} \right\rangle = a + bE, \qquad a = 2.0 \times 10^{-3}\ \mathrm{GeV\,g^{-1}cm^2}, \quad b = 4.0 \times 10^{-6}\ \mathrm{g^{-1}cm^2}, \tag{4}$$

where X is opacity. Solving $dE/dX = -(a + bE)$ gives the exact minimum survival energy

$$E_{\min}(\varrho) = \frac{a}{b}\left(e^{b\varrho} - 1\right). \tag{5}$$

For small $b\varrho$ this reduces to $E_{\min} \approx a\varrho$. At $\varrho = 25{,}000$ $\mathrm{g\,cm}^{-2}$ (100 m of limestone), Eq. (5) gives 52.6 GeV, 5.2% above the linear estimate of 50 GeV.

The expected number of muons in one pixel is

$$\mu = \Phi(E_{\min}(\varrho), \theta)\, A\, \Omega\, T, \tag{6}$$

where Φ is the integrated muon flux above $E_{\min}$, A is the detector area, T is the exposure time, and Ω is the pixel solid angle. The flux uses the Gaisser parametrisation [8] with the low-energy correction of Guan et al. [9]. The project uses the small-angle convention $\Omega \approx (\Delta\theta)^2$, with $\Delta\theta = 2^\circ = 3.49 \times 10^{-2}$ rad. At the signal direction (about 30° from vertical) this overestimates the exact tangent-plane solid angle by about 13%, and at the edge of the field by about 23%. The effect is treated as a systematic in Section 5.

Two maps are always produced, with one for the target with the chamber and one for the twin target with the chamber filled solid. Their ratio

$$R = \frac{\mu_{\text{void}}}{\mu_{\text{rock}}} \tag{7}$$

is the reconstructed radiograph. Away from the chamber, $R = 1$ because the pyramid's shape cancels. However, over the chamber, $R > 1$ because the missing rock transmits extra muons.

2.4 Statistical model

A simulated observation is a Poisson draw from μ. The observed significance of a count N against a no-chamber expectation μ_0 is

$$S_{\text{obs}} = \frac{N - \mu_0}{\sqrt{\mu_0}}, \tag{8}$$

which changes with the random seed. The expected significance compares two expectation maps:

$$S_{\text{exp}} = \frac{\mu_{\text{void}} - \mu_{\text{rock}}}{\sqrt{\mu_{\text{rock}}}}. \tag{9}$$

This is deterministic for fixed parameters and defines the smooth exposure curve. Because both the signal and the background scale linearly with T, $S_{\text{exp}} \propto \sqrt{T}$.

The official headline uses S_{exp} at a single signal pixel chosen once from the noiseless maps, at $(a_x, a_y) = (0°, 30°)$. The line of sight to the chamber centre is at $\tan\theta = 40/(70 - 3) = 0.597$, or $\theta = 30.8°$. Choosing the brightest pixel in each noisy map would select upward fluctuations and bias the result. The image has 1681 pixels, so a noise-only image is expected to contain about $1681 \times 1.35 \times 10^{-3} = 2.27$ pixels above 3σ. This look-elsewhere effect is quantified in Section 5, and the 5σ level is the project's predefined benchmark.

The main methodological limitations are that the model is a two-dimensional radiograph rather than a three-dimensional tomographic inversion, that muon scattering and detector response are not simulated, and that the rock density is assumed known exactly. These choices are justified by the project's scope and are revisited in Section 6.

3 Data Collection

3.1 Source and structure of the data

The data in this project are synthetic by design, as they are generated by the simulation rather than measured with a physical detector. This is appropriate for a controlled study of the exposure question, because the true density distribution is known exactly and the same geometry can be run with and without the chamber. It also means that the results test the internal consistency of the model, not the performance of a real instrument.

Five data products were generated. (i) Two 1681-pixel expectation maps at 1 and 30 days, one with the chamber and one without, from which the clean ratio and the expected significance are computed. (ii) A primary Poisson observation at 1 and 30 days using reference seed 0. (iii) A null ensemble of 500 independent 30-day observations of the solid pyramid, used to characterise the look-elsewhere effect. (iv) An ensemble of 500 observations of the chamber target at 7, 14 and 30 days, used to characterise the fluctuation of a real measurement around the expected curve. (v) One-parameter scans, which include chamber radius (4–20 m), chamber depth (55–110 m), detector distance and lateral offset, angular bin size (1–5°), detector area (0.0625–2 m^2), and assumed rock density (2.3–2.7 g cm^{-3}). A hypothetical Malaysian limestone hill provides an additional target.

Each pixel represents one viewing direction, specified by the two pointing angles a_x and a_y in degrees. The primary arrays are 41×41, and the $1°$ binning scan uses 81×81. Counts are stored as

floating-point expectations before sampling and as integer Poisson draws afterwards. Opacities are in $\mathrm{g\,cm^{-2}}$, densities in $\mathrm{g\,cm^{-3}}$, angles in degrees internally and radians in the flux functions, and exposure times in days. All arrays use a common orientation, `[iy, ix]`, so a map can be plotted with the angle grids directly.

3.2 Preparation and bias controls

No smoothing, interpolation, outlier removal or missing-value imputation is applied, as the simulation produces a complete array by construction. The only pre-processing step is the choice of the signal pixel, which is made once from the noiseless 1-day expectation maps and then held fixed for every exposure and every noise realisation. This avoids the selection bias that would arise from choosing the brightest pixel in each noisy map. The reference seed is fixed at 0, and every noisy figure states it. Where an ensemble is used, the ensemble mean and standard deviation are reported rather than a single draw.

The main sources of bias are model assumptions rather than data handling. The detector is treated as a point inside the target footprint, so the rock between the assumed niche and the outside is not counted. The 2° pixels are perfectly independent, whereas a real detector has finite angular resolution. The rock density is assumed to be exactly $2.5\ \mathrm{g\,cm^{-3}}$. These assumptions are quantified in Section 5 and discussed in Section 6. The data are representative only of the stated model, and they are not a sample of a real pyramid.

3.3 Data and code availability

The data are reproducible from the notebooks in the repository. A summary of every number used in this paper is written to `figures/results_summary.json`. The repository contains the simulation code, the analysis module, the notebooks, and all figures. Appendix A lists the data sources and software, whereas Appendix D maps figures to notebooks.

4 Analysis

4.1 Analytical framework and implementation

The core analysis chain includes the opacity map $\rightarrow$ minimum energy $\rightarrow$ expected counts $\rightarrow$ ratio map $\rightarrow$ expected significance $\rightarrow$ exposure curve. The two integral functions are `opacity_map`, which builds the angular grids and traces one ray per pixel, and `expected_counts`, which applies Eqs. (6)–(7). A third helper, `counts_from_opacity`, reuses a traced geometry for exposure scans, and because counts are exactly proportional to T, this introduces no approximation. The functions are imported by notebooks 03-05, so every notebook runs from a fresh kernel without relying on variables left in memory.

The expected-significance curve is evaluated at 14 exposures from 1 to 180 days. The 5σ crossing is obtained by linear interpolation between the two bracketing points, and the exact square-root scaling gives a second, independent estimate. The four parameter studies change one variable at a time. Studies A–C evaluate the signal along the exact detector-to-chamber-centre ray rather than at the maximum of a coarse map, which removes a grid-alignment artefact that otherwise makes the curves jagged at the few-per-cent level. Study D uses the full map because the angular binning is the variable under study. Log–log fits are used only to describe the scaling of significance and exposure with chamber radius.

Uncertainty is handled at two levels. Statistical uncertainty is estimated from the 500-member ensembles, and include the standard deviation of the observed significance at a fixed pixel, and the distribution of the maximum pixel in the null experiment. Systematic uncertainty is estimated by repeating the baseline analysis with (i) the exact tangent-plane pixel solid angle, (ii) a range of assumed rock densities, (iii) a range of detector areas, and (iv) a per-pixel versus whole-image definition of significance. The headline itself has no statistical error bar, as it is a deterministic property

Table 2. Validation checks. The reference is an independent hand calculation, a control target, or a statistical expectation.

Check	Code result	Reference	Agreement
Integrated flux above 1 GeV	60.14	≈ 60 (Gaisser/PDG)	0.2%
$E_{\min}$ at 100 m limestone	52.59 GeV	52.59 GeV (exact)	exact
$E_{\min}$ vs linear rule	52.59 GeV	50 GeV (linear)	5.2%
Central opacity, vertical ray	21,999	21,913 (geometry)	0.4%
Chamber opacity deficit	4,935	5,000 ($2r\rho$)	1.3%
Ray step 0.25 $\rightarrow$ 0.125 m	—	—	max 0.24%
No-void control	0.00	0 by construction	exact
Clean ratio away from chamber	1.0000	1 by construction	exact
Poisson spread at mean 10,000	100.9	$\sqrt{10{,}000} = 100$	0.9%
Null maximum (500 runs)	3.44 mean	look-elsewhere	see text

of the expectation maps.

4.2 Validation

Table 2 lists the checks. The two most important are the ray step convergence and the no-void control. Halving the step from 0.25 m to 0.125 m changes the opacity map by at most 0.24% and by 0.07% on average. The no-void control is zero to machine precision away from the chamber, which shows that the difference map isolates the chamber rather than a geometry bug.

At the chamber the opacity deficit is $4{,}935$ g cm^{-2}, against $2r\rho \times 100 = 5{,}000$ for a 20 m chord, and the 1.3% difference comes from the discrete ray missing the centre by about 1.3 m. The central vertical opacity is 21,999, against 21,913 from the exact pyramid geometry, and the 0.4% difference is the 0.25 m marching step. Drawing 5,000 Poisson samples at means 10, 100, 1,000 and 10,000 gives standard deviations 3.2, 9.9, 31.7 and 100.9, compared with $\sqrt{\mu} = 3.16$, 10, 31.6 and 100. The fixed-pixel null distribution has mean zero and standard deviation about one.

5 Results

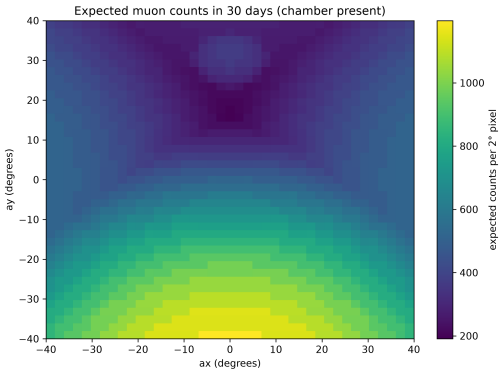
5.1 Target and opacity

Figure 6 in the Appendix shows the density slice and the opacity difference produced by the chamber. The raw opacity ranges from $14{,}315$ to $34{,}993$ g cm^{-2}, a factor 2.4, because different lines of sight cross different amounts of the pyramid. The chamber removes $4{,}935$ g cm^{-2} along the line of sight, about 19% of the local opacity and within 1.3% of the geometric expectation $2r\rho \times 100 = 5{,}000$ for a 20 m chord. The no-void control is flat to machine precision away from the chamber. These two results are the first evidence that the ray tracer and the density model are correct.

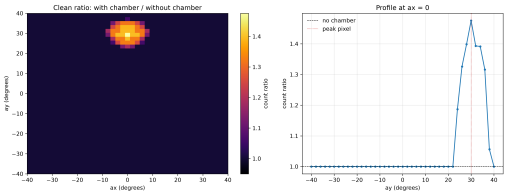
5.2 Expected counts and the before/after pair

Figure 1 shows the 30-day expected-count map and the clean ratio map. The raw counts run from 191 to 1,197 per pixel, a factor of 6.3, and the total is 9.85×10^5 muons. The chamber changes the total by only 0.36%, because it affects a small fraction of the image. The raw map is therefore a picture of the pyramid, not of the chamber.

The clean ratio is 1.0000 away from the chamber and peaks at 1.4758 at $(a_x, a_y) = (0^\circ, 30^\circ)$. At that pixel the expected counts are 366.8 with the chamber and 248.5 without, an excess of 118.3 muons in 30 days, or $S_{\exp} = 7.50\sigma$. The chamber is recovered from the muon data alone, using the no-void twin as the reference. This ratio map is the after half of the before/after pair, as the density slice is the before half.



(a) Expected counts in 30 days. The pyramid dominates.



(b) Clean ratio with the no-void twin, and a profile at $a_x = 0°$. The chamber is the excess above 1.0.

Figure 1. The before/after pair. The left panel is what the detector counts, and the right panel is the reconstructed radiograph after dividing out the known shape of the pyramid.

5.3 Noise, observed significance, and the look-elsewhere effect

Figure 5 in the Appendix shows the same analysis after Poisson sampling with seed 0. At one day the image is noise-dominated, as the largest observed significance is 4.07σ somewhere by chance, while the fixed signal pixel is only 1.29σ. At 30 days the chamber is a visible cluster, and the fixed pixel has observed significance 7.70σ, close to the expected 7.50σ. The maximum pixel also happens to be 7.70σ because the signal is real, as at one day the maximum and the fixed pixel are unrelated.

The null control is more informative. In 500 independent 30-day simulations of the solid pyramid, the largest pixel has mean 3.44σ, median 3.40σ, and 95th percentile 4.12σ. It exceeds 3σ in 91.6% of images and 5σ in none. The mean number of pixels above 3σ is 2.68, close to the naive expectation $1681 \times 1.35 \times 10^{-3} = 2.27$. A single 3σ pixel in a 1681-pixel image is therefore not interesting. This is the look-elsewhere effect, and it is the reason the headline uses a pre-selected pixel and the smooth expected curve. The full null distributions are shown in Appendix Figure 10.

5.4 Exposure curve and the headline result

The expected significance at the fixed pixel is 1.3696 after one day. Because $S_{\text{exp}} \propto \sqrt{T}$, the curve in Figure 2 follows $S(T) = 1.3696\sqrt{T/\text{day}}$. It crosses 3σ at 4.82 days and 5σ at 13.37 days by linear interpolation. Meanwhile, the exact square-root solution is 13.33 days. We therefore quote that

> **For the baseline 0.25 m^2 detector and a 10 m radius chamber, the expected chamber signal reaches 5σ after approximately 13.4 days under the project's simplified model.**

There is no statistical uncertainty on this number as it is a deterministic property of the model. The red points in Figure 2 show what a sequence of measurements would look like. At 7 days the mean observed significance is 3.75 ± 1.19, and 15% of experiments cross 5σ, at 14 days the mean is 5.12 ± 1.24, and 54% cross, and at 30 days the mean is 7.53 ± 1.23, and 98% cross. The expected curve is the average of these distributions, not a prediction of any one of them. The individual distributions are shown in Appendix Figure 7b.

5.5 Parameter studies

All four named studies were completed. Table 3 summarises them and Figure 3 shows the results.

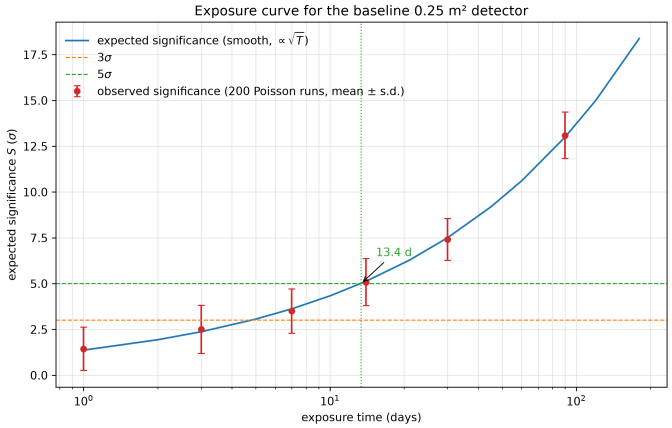


Figure 2. Expected significance versus exposure, with 3σ and 5σ reference lines. The smooth curve is deterministic, and the red points are the mean and standard deviation of 200 Poisson experiments at selected exposures. The 5σ crossing is 13.4 days.

Table 3. Summary of the four named parameter studies. Each study changes one parameter, and all others remain at the baseline.

Study	Range	Result
A: radius	4–20 m	T_5 146.6 $\rightarrow$ 1.99 days. $S \propto r^{1.32}$
B: depth	55–110 m	T_5 15.4 $\rightarrow$ 21.6 $\rightarrow$ 17.8 days
C: placement	$y = -55$ to -15 m	optimum $y \approx -25$ m, 15.9 days
D: bin size	1°–5°	peak S 3.75 $\rightarrow$ 18.75. image $\approx$ 3.5

Study A: chamber size. Varying the radius from 4 to 20 m at fixed depth and 30 days gives significance from 2.26σ to 19.43σ and exposure from 146.6 to 1.99 days. A log–log fit gives $S \propto r^{1.32}$ and $T_5 \propto r^{-2.65}$. The exponent is close to one as the radiograph measures the missing chord through the chamber, which grows as r, not the missing volume, which grows as r^3.

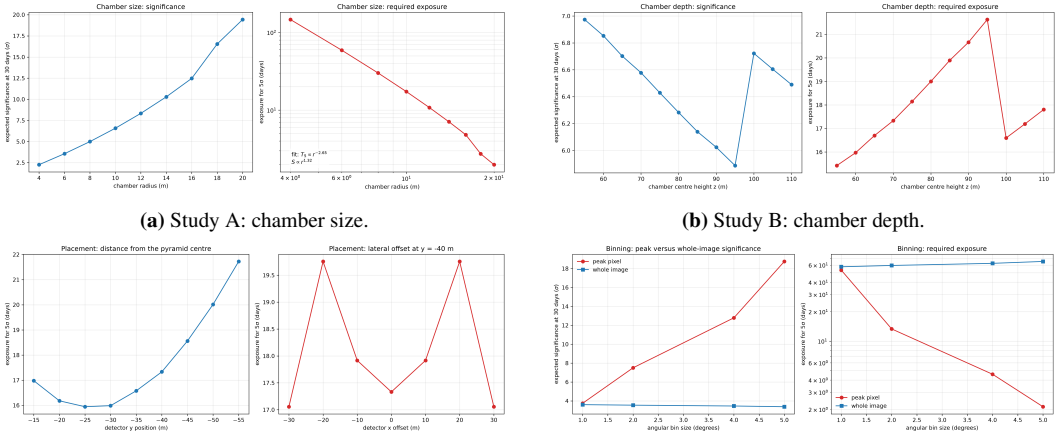
Study B: chamber depth. The accessible range is 55-110 m, as below about 52 m the line of sight is outside the $\pm 40°$ field. The exposure is 15.4 days at 55 m, rises to 21.6 days at 95 m, and falls to 17.8 days at 110 m. The shallow end has less rock but a larger zenith angle, whereas the deep end has more rock but a more vertical flux. Above 95 m, the chamber approaches the apex, the ray exits the pyramid soon after the chamber, and the overburden stops growing.

Study C: detector placement. The distance scan has an optimum near $y = -25$ m, where the exposure is 15.9 days, against 17.3 days at the baseline $y = -40$ m and 21.7 days at $y = -55$ m. The lateral response is flat within ± 30 m (17.1–19.8 days), with a weak non-monotonicity caused by the line of sight exiting through different faces of the pyramid.

Study D: angular binning. The peak-pixel significance rises from 3.75σ at 1° to 7.50σ at 2°, 12.79σ at 4° and 18.75σ at 5°. This is arithmetic, not physics, as the counts per pixel scale with $\Omega \propto (\Delta\theta)^2$, so $S_{\mathrm{peak}} \propto \Omega/\sqrt{\Omega} \propto \Delta\theta$. The whole-image significance is almost constant at 3.39–3.61σ. The coarse-bin gain is a change in the statistic, not a real gain in detectability. A real detector would also blur neighbouring directions, which this model does not include.

5.6 Additional targets and systematics

Table 4 collects the alternative exposure estimates. Using the exact tangent-plane pixel solid angle instead of the project's small-angle convention increases the 5σ exposure from 13.3 to 15.4 days. The whole-image statistic, which integrates the excess over all 1681 pixels and divides by the square root of the total background, gives 3.55σ at 30 days and a 5σ exposure of 59.5 days. Neither is a



(a) Study A: chamber size.

(b) Study B: chamber depth.

(c) Study C: detector placement.

(d) Study D: angular binning.

Figure 3. The four named parameter studies. Each panel varies one parameter at a time and makes one point.

Table 4. Summary of 5σ exposure estimates under different definitions and assumptions. The official value is shown in bold.

Definition or variation	Note	T_5 (days)
Chamber radius 20 m	Study A	2.0
Baseline, peak pixel	**official**	**13.3**
Baseline, exact pixel solid angle	systematic	15.4
Baseline, whole image (global)	conservative	59.5
Hypothetical Malaysian hill	8 m cave	9.1
Chamber radius 4 m	Study A	146.6
Detector area 0.0625 m^2	scaling	53.3

contradiction, as the per-pixel statistic asks whether one pre-selected direction is bright, while the whole-image statistic asks whether the total excess is significant.

Detector area scales exactly as expected, which is $T_5 \propto A^{-1.00}$, from 53.3 days at 0.0625 m^2 to 1.67 days at 2 m^2 (Appendix Figure 7a. The observed fluctuation is in Figure 7b). The hypothetical Malaysian limestone hill, with an 8 m cave seen through about 80 m of rock, reaches 5σ in 9.1 days (Appendix Figure 8a). The most important systematic is the assumed rock density (Appendix Figure 9), as analysing a solid target with 2.6 instead of 2.5 g cm^{-3} produces a fake maximum of 3.86σ at 30 days and 9.46σ at 180 days, and at the chamber pixel the fake signal is 1.11σ and 2.72σ. With 2.7 g cm^{-3} the 180-day maximum is 14.76σ and the chamber pixel is 5.36σ. A void and slightly lighter rock are degenerate in a line-integrated measurement. All estimates are collected in Table 4 and Appendix Figure 11.

6 Discussion

The headline is a deterministic benchmark, as with the baseline parameters, the expected signal at one pre-selected direction crosses 5σ at 13.4 days. It is not a measurement or a prediction for Khufu. Its value is that the geometry, the spectrum, the energy-loss model, the Poisson sampling, and the choice of statistic are visible. The result is therefore reproducible and can be used to teach why exposure, angular resolution, and systematics matter.

The result is model-dependent in four important ways. First, detector acceptance and efficiency are absent, as the simulation counts every muon in every pixel. A 50% efficient detector doubles the exposure. Second, backgrounds are absent, thus electrons from local radioactivity and muons

scattered in from outside the target would add counts that are not modelled. Third, the atmospheric muon flux varies with pressure and temperature, thus a real campaign must monitor and correct it. Fourth, and most important, the rock density is assumed known. A 4% error already produces a fake 3.86σ maximum at 30 days and a fake 9.46σ maximum at 180 days. Muography measures density along a line, so a void and slightly lighter rock are degenerate unless an independent density constraint is available.

The look-elsewhere effect is a second, statistical limitation. A solid pyramid produces a 3σ pixel in 91.6% of 30-day images because the image has 1681 pixels. The project handles this by fixing the signal pixel from the noiseless model and by reporting the null distribution. A real analysis would go further and quote a global p-value or a trials factor. The 5σ level used here is a project benchmark, not a discovery criterion.

The comparison with the 2017 Khufu measurement is at the level of physics and scale, not a reproduction of the analysis. The ScanPyramids collaboration found a large void above the Grand Gallery with three independent detector technologies [4]. Their measurement faced the same basic physics as this simulation, but also detector acceptance, efficiency, angular resolution, backgrounds, and a real, non-uniform target. A roughly 10 m void inside a roughly 100 m rock target is a several-σ measurement only with a long exposure and careful control of systematics, exactly as the 2017 result showed.

The limitations of this study are clear. It is a two-dimensional radiograph, not a three-dimensional tomographic inversion. Muon scattering, detector response, backgrounds, and atmospheric variation are not simulated. The detector is an idealised point inside the target footprint, the pixel solid angle uses a small-angle approximation, and the rock density is fixed. The study also uses synthetic data by design, so it tests the consistency of the model rather than the performance of a real instrument.

Several specific next steps follow. A detector-response model with efficiency, angular resolution, and a point-spread function would show how the chamber signal is diluted by real binning. A Monte Carlo over rock density and detector efficiency would turn the single-point systematics into a confidence interval. A full 3D inversion using several detector positions would move from a radiograph to a density model. Finally, a real digital elevation model of a Malaysian limestone hill would replace the conical toy target and make the stretch estimate directly useful.

7 Conclusion

We built a reproducible end-to-end simulation of muon radiography for a 230 m limestone pyramid with a 10 m radius hidden chamber, including ray tracing, expected counts, Poisson noise, significance, null controls, and four parameter studies. Under the baseline model the expected chamber signal reaches 3σ after 4.8 days and 5σ after 13.4 days, and the 30-day expected significance is 7.50σ. The answer is controlled by geometry and statistics rather than by the muon flux alone, as the signal follows the missing chord, the look-elsewhere effect makes a single 3σ pixel unsurprising, and a 4% density error can fake a 5σ signal at long exposure. The main limitations are the absence of detector response, scattering, backgrounds, atmospheric variation, and a full 3D inversion. The most useful next steps are a detector-response model, a density Monte Carlo, and a tomographic inversion with multiple detector positions.

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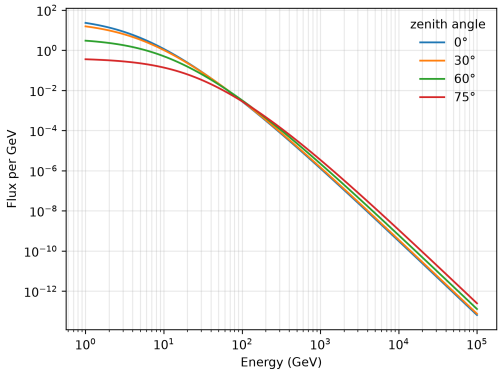
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Appendix

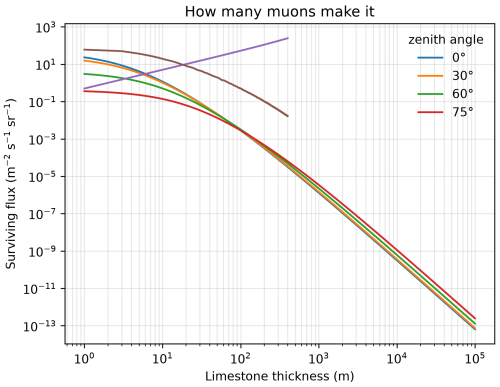
A Data Sources and Software

This project uses synthetic data generated by the simulation, and no external observational dataset was used. The primary inputs are the baseline parameters in `code/project_config.py`, the supplied physics in `code/muography.py`, and the target definition in the notebooks. The software environment is Python 3.11.17 with NumPy 2.4.6 [5], Matplotlib 3.11.2 [6], and Jupyter 1.1.1 [7]. The reference seed is 0. All generated data are reproducible by running notebooks 02–05 in order, and the summary of every number used in this paper is in `figures/results_summary.json`.

B Additional Figures



(a) Differential muon flux at four zenith angles. The spectrum is harder at large angles because low-energy muons are removed by the longer atmospheric path.



(b) Transmitted flux above the minimum energy through increasing limestone thickness. The vertical scale is logarithmic.

Figure 4. The two physics ingredients used by the simulation. Both panels are produced by `notebooks/01_muons_and_rock.ipynb`.

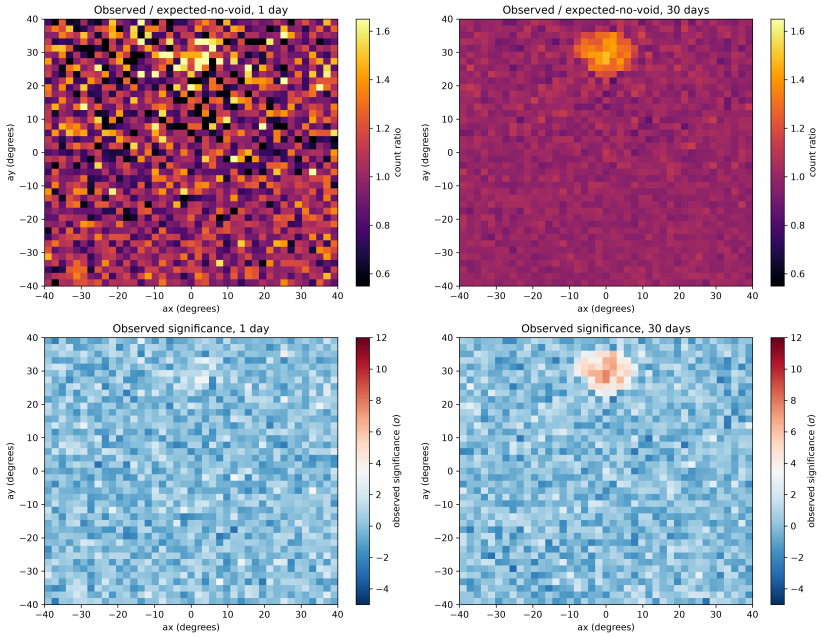
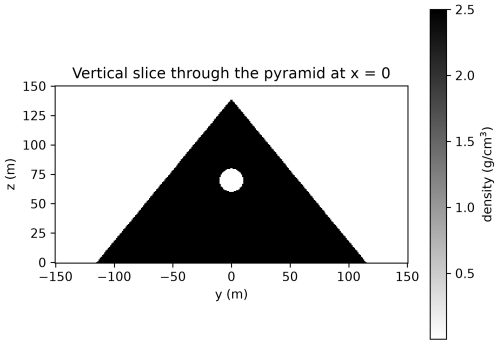
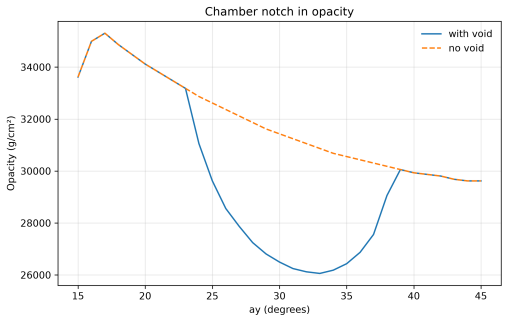


Figure 5. Poisson-noisy observations at 1 and 30 days. Top: observed counts divided by the no-void expectation. Bottom: observed significance. The chamber is invisible at one day and clear at 30 days. Seed 0.

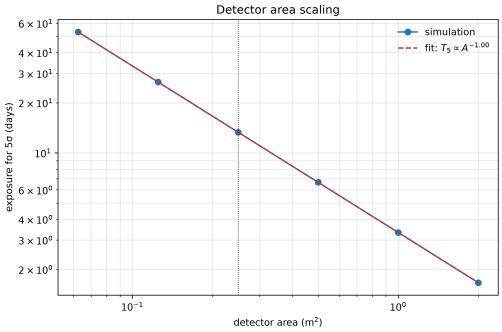


(a) Density slice at $x = 0$. The chamber is the circular air-filled region at $z = 70$ m.

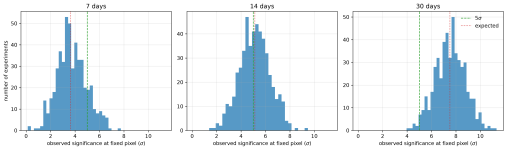


(b) Opacity versus vertical pointing angle at $a_x = 0°$ with and without the chamber. The notch is the chamber signature.

Figure 6. The target and its first ray-traced signature. The chamber lowers the opacity by about $4{,}900\ \mathrm{g\,cm^{-2}}$. The no-void control is flat.

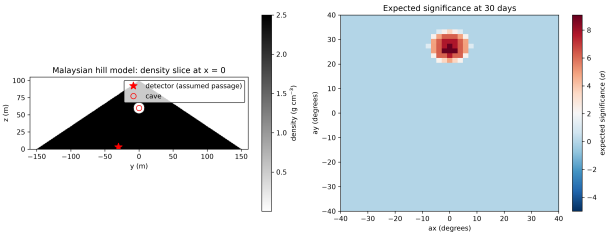


(a) Detector-area scaling. The fit is $T_5 \propto A^{-1.00}$.

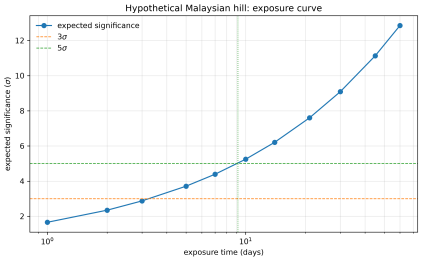


(b) Observed significance at the fixed pixel for 7, 14 and 30 days, 500 experiments each.

Figure 7. Supplementary checks: detector-area scaling and the fluctuation of a real observation around the expected curve.



(a) Hypothetical Malaysian hill: density slice and expected significance at 30 days.



(b) Exposure curve for the same hill, 5σ at 9.1 days.

Figure 8. The hypothetical Malaysian limestone-hill extension. This is a toy target, not a survey of a real site.

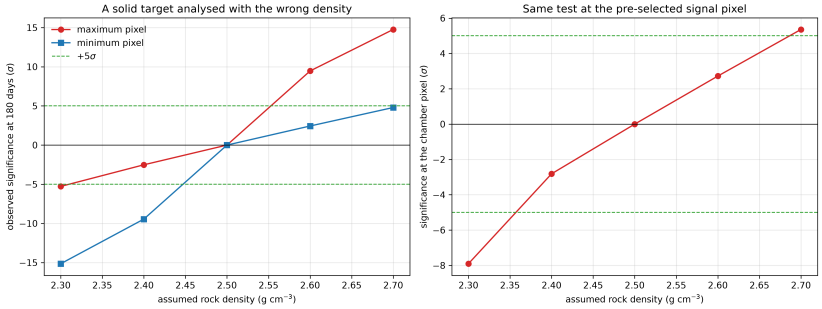


Figure 9. Density systematics. A solid target is analysed with the wrong assumed density. A 4% error produces a fake maximum above 5σ by 180 days, and the fake signal follows the shape of the target.

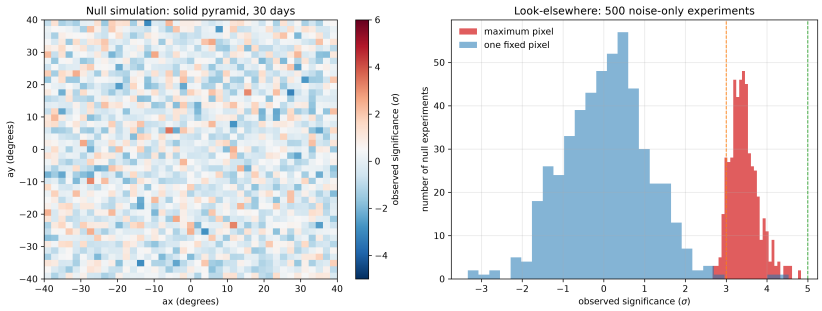


Figure 10. The no-void control. Left: one noise-only significance map. Right: the distributions of the maximum pixel and one fixed pixel over 500 null experiments. The maximum is pushed to $\approx 3.4\sigma$ by the number of pixels searched.

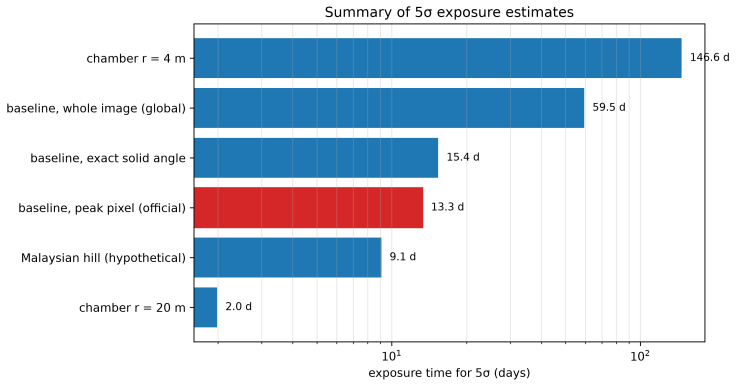


Figure 11. Summary of the 5σ exposure estimates under different definitions and assumptions. The official baseline is highlighted.

C Derivation of the Minimum Survival Energy

Starting from

$$\frac{dE}{dX} = -(a + bE), \tag{10}$$

separate variables and integrate from $E = E_{\min}$ at $X = 0$ to $E = 0$ at $X = \varrho$:

$$\int_{E_{\min}}^{0} \frac{dE}{a + bE} = -\int_{0}^{\varrho} dX. \tag{11}$$

With $u = a + bE$, $du = b\,dE$, the left side is

$$\frac{1}{b} \ln\left(\frac{a}{a + bE_{\min}}\right) = -\varrho, \tag{12}$$

so

$$E_{\min}(\varrho) = \frac{a}{b}\left(e^{b\varrho} - 1\right). \tag{13}$$

For small $b\varrho$, $e^{b\varrho} - 1 \approx b\varrho$ and $E_{\min} \approx a\varrho$. At $\varrho = 25{,}000$ g cm^{-2} the exact value is 52.59 GeV, 5.2% above the linear 50 GeV.

D Code Availability

The repository is `muography-project/`. The figure-to-notebook mapping is given in Table 5. The shared analysis functions are in `code/student_analysis.py`, the baseline parameters are in `code/project_config.py`, and the summary of all numbers is `figures/results_summary.json`. Notebooks 03–05 import the shared module and run from a fresh kernel.

Table 5. Figure-to-notebook mapping.

Figure(s)	Notebook
Fig. 4	`01_muons_and_rock.ipynb`
Fig. 6	`02_ray_tracing.ipynb`
Fig. 1	`03_first_image.ipynb`
Figs. 5, 2	`04_noise_and_exposure.ipynb`
Figs. 3 and Appendix figures	`05_parameter_study.ipynb`